
DistServe & TaiChi

INTRODUCTION TO LLM
INFERENCE SERVING SYSTEMS
CHUHONG YUAN





Outline

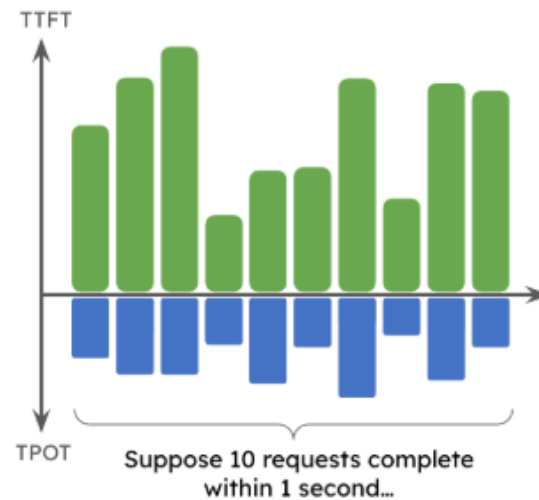
- Goodput vs. Throughput
- DistServe
- TaiChi
- Discussion: limitations of P-D disaggregation
- Homework review

Goodput vs. Throughput

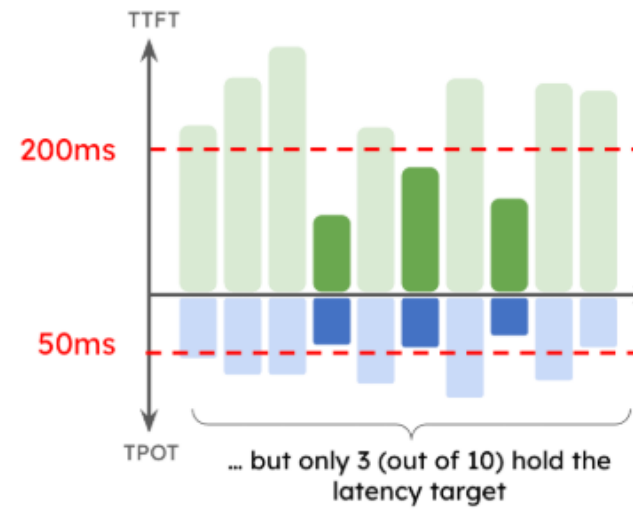
Throughput Is Not All You Need

High Throughput $\neq$ High Goodput

Throughput = completed request / time
= 10 req / s

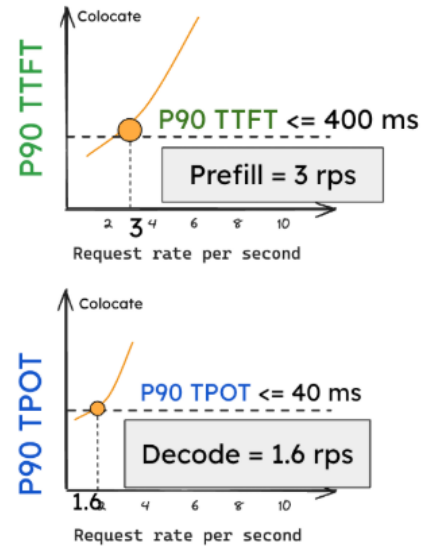


Goodput = completed requests **within SLO** / time
= 3 req / s



Arguments – Better RPS In Disaggregated Serving

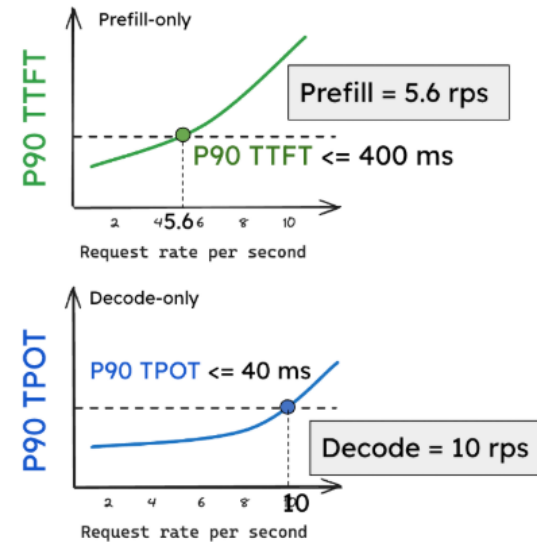
Colocate



 Max System rps
= $\text{Min}(\text{Prefill}, \text{Decode})$
= 1.6 rps / GPU

Disaggregate (2P1D)

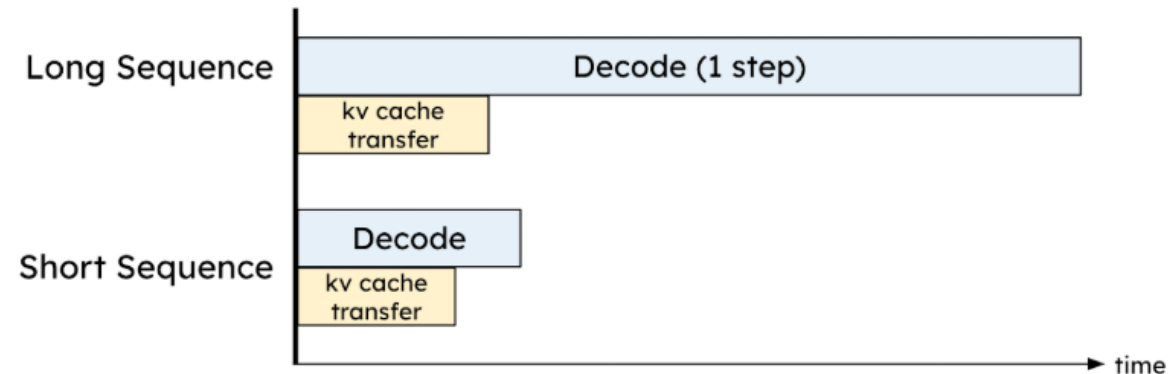
2 GPU for Prefill + 1 GPU for Decode



  Disaggregate (2P1D)
= $\text{Min}(5.6 \times 2, 10)$ rps / 3 GPU
= 3.3 rps / GPU

Arguments – KV Cache Transfer Is Not Dominant

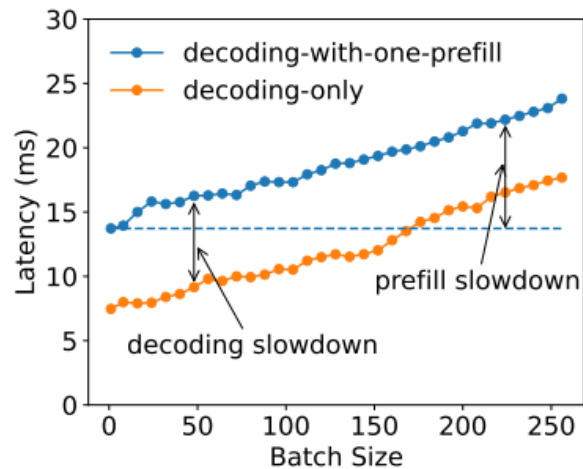
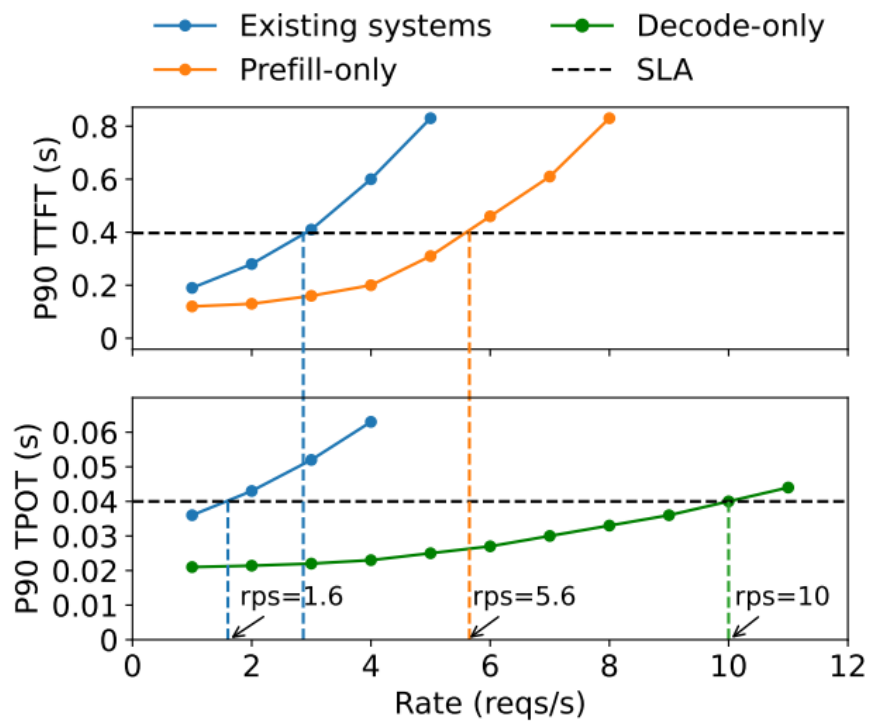
- 8 channel, 64GB/s per link
- One request, 2048 tokens, 4.5 MB/token
- Latency = $2048 \text{ tokens} * (4.5 \text{ MB/token}) / (64\text{GB/s} * 8) = 17.6 \text{ ms}$



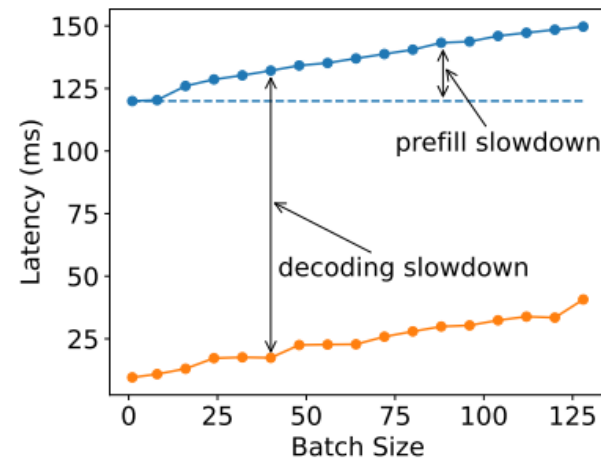
DistServe



Motivation

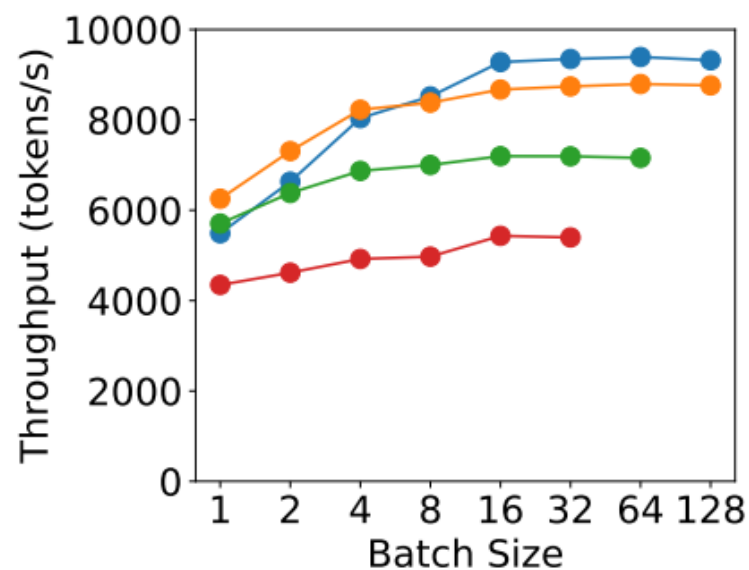


(a) Input length = 128



(b) Input length = 1024

Analysis - Prefill



(a) Prefill phase

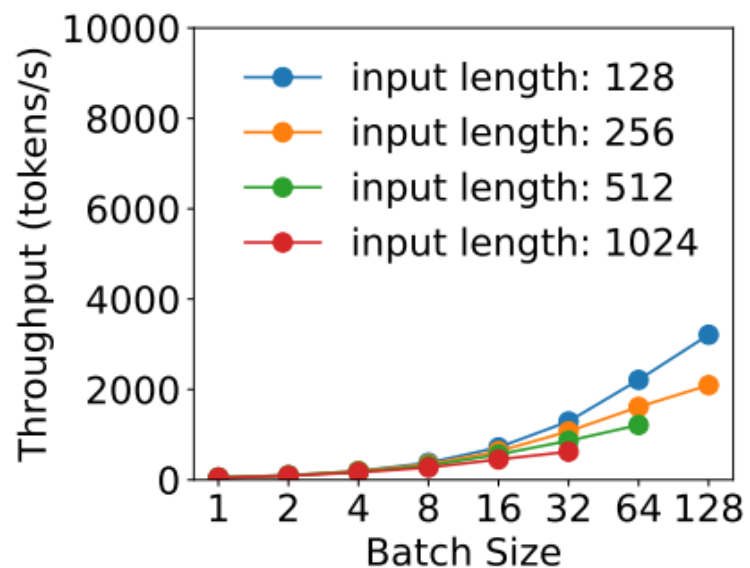
- Inter-op

$$Avg_TTFT_{inter} = D_s + \frac{RD_m^2}{2(1 - RD_m)} = D + \frac{RD^2}{4(2 - RD)}.$$

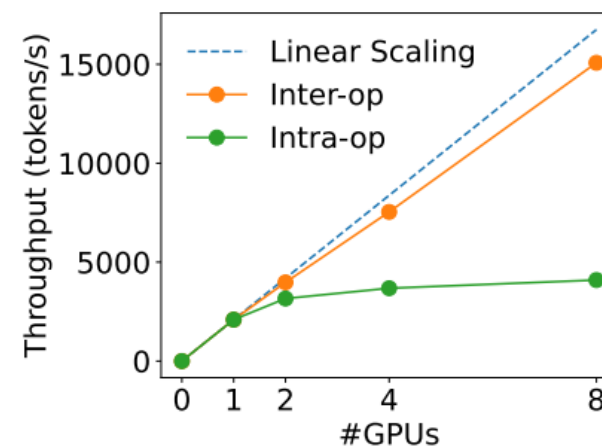
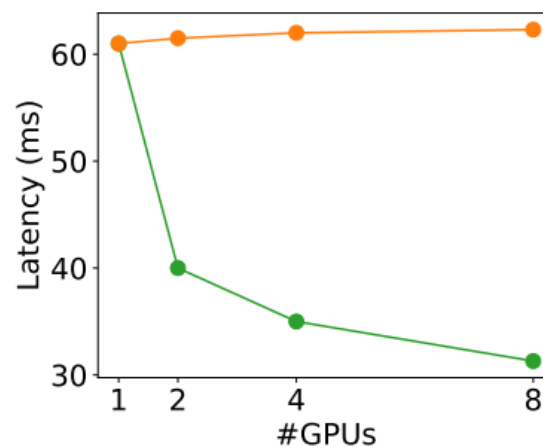
- Intra-op

$$Avg_TTFT_{intra} = \frac{D}{K} + \frac{RD^2}{2K(K - RD)}.$$

Analysis - Decoding



(b) Decoding phase





Method

- High node-affinity cluster
 - Within the memory capacity
 - For each parallelism option, use the simulator to predict the best goodput, and select the best one
- Low node-affinity cluster
 - Put prefill and decoding of the same stage in the same node
 - Enumerate each node's configuration to find the best one

Evaluation – Settings

- 4 × 8 A100 80GB

Application	Model Size	TTFT	TPOT	Dataset
Chatbot OPT-13B	26GB	0.25s	0.1s	ShareGPT [8]
Chatbot OPT-66B	132GB	2.5s	0.15s	ShareGPT [8]
Chatbot OPT-175B	350GB	4.0s	0.2s	ShareGPT [8]
Code Completion OPT-66B	132GB	0.125s	0.2s	HumanEval [14]
Summarization OPT-66B	132GB	15s	0.15s	LongBench [13]

Table 1: Workloads in evaluation and latency requirements.

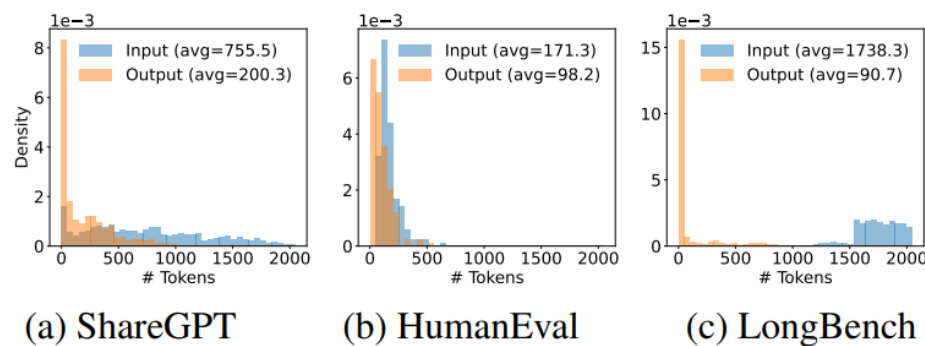


Figure 7: The input and output length distributions of (a) ShareGPT, (b) HumanEval, and (c) LongBench datasets.

Evaluation - Chatbot

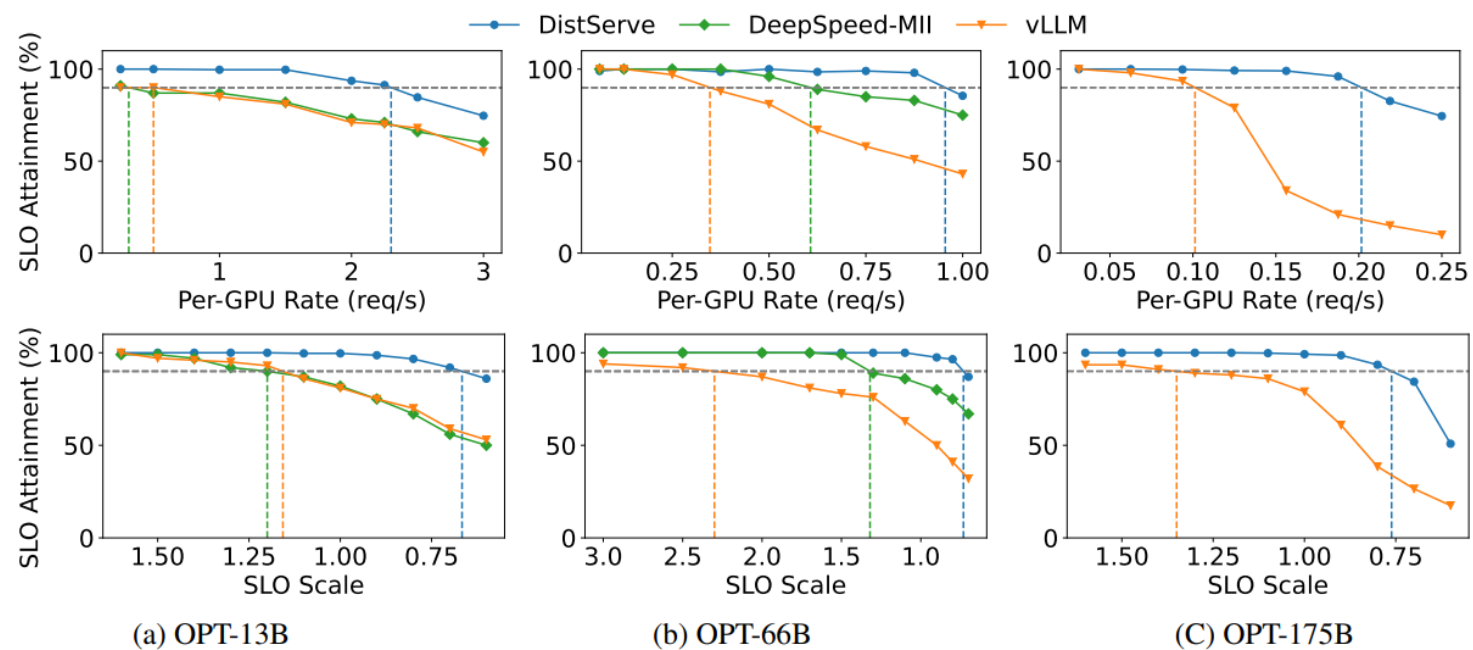


Figure 8: Chatbot application with OPT models on the ShareGPT dataset.

Evaluation – Code Completion & Summarization

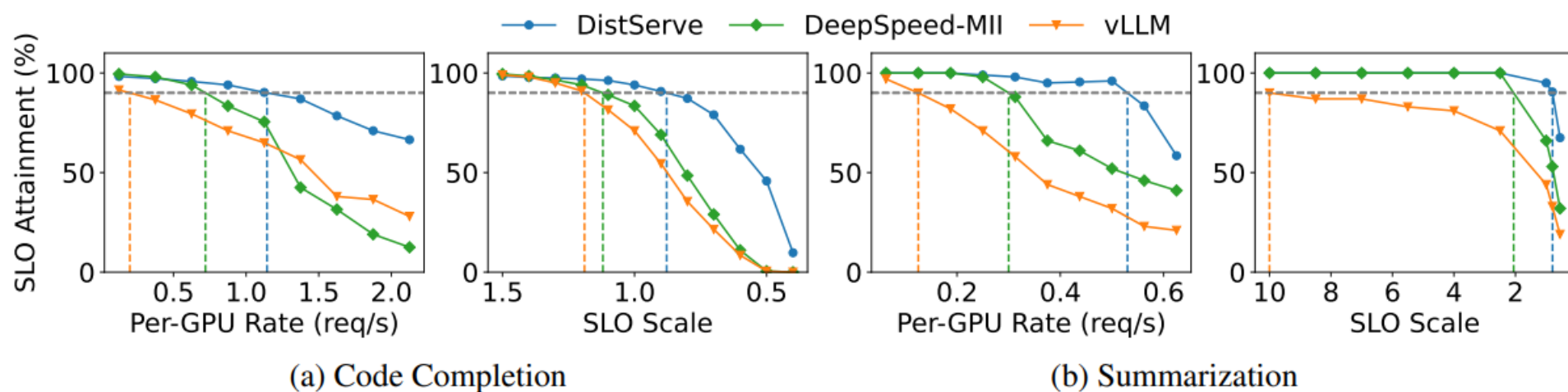
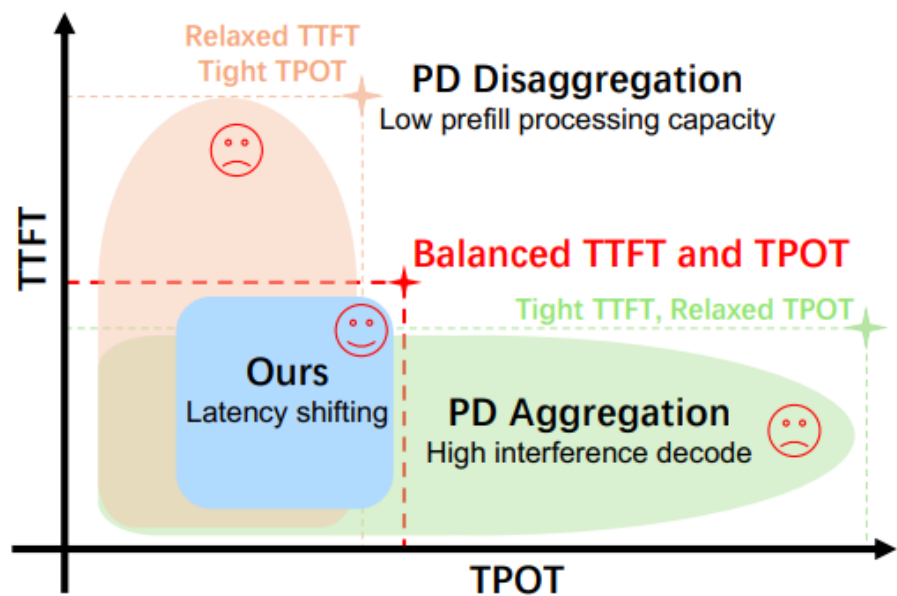


Figure 9: Code completion and summarization tasks with OPT-66B on HumanEval and LongBench datasets, respectively.

TaiChi



Motivation



Motivation – Colocated Serving

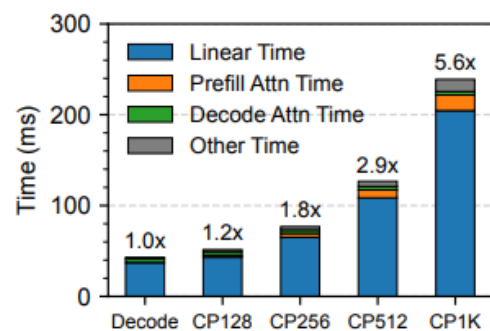
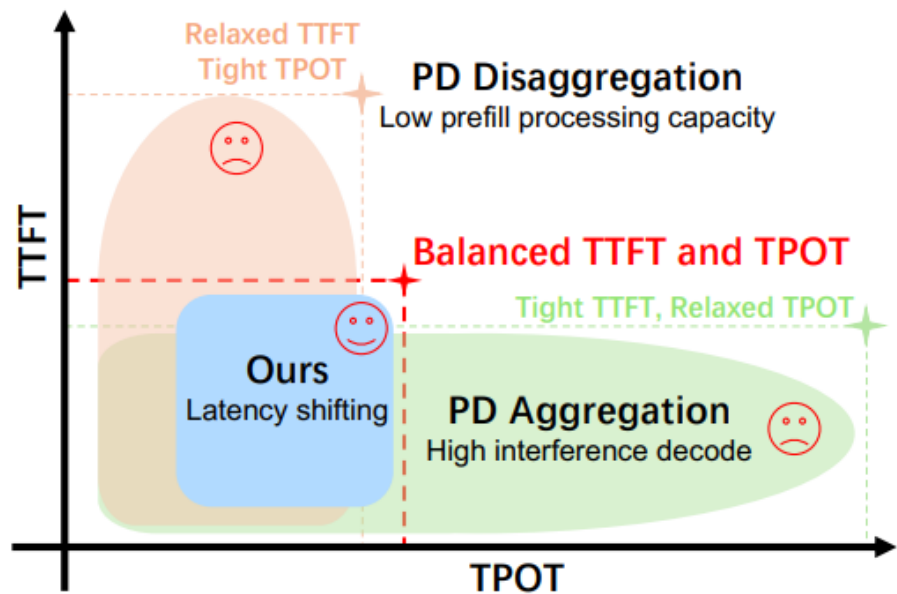


Figure 3. Breakdown of batch execution time with varying chunk sizes (batch size = 16). *CPxxx* refers to Chunked Prefill with a chunk size of xxx.

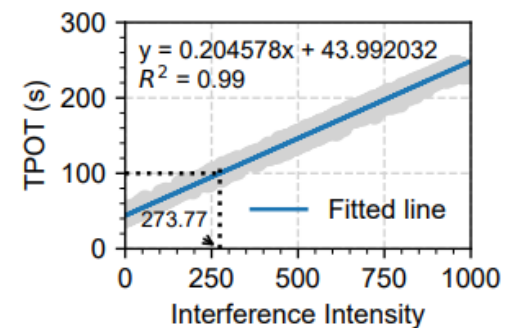


Figure 4. Scatter plot of the requests' TPOT and their suffered interference intensity (CP1024), with a fitted line ($R^2 = 0.99$, strong correlation).

Motivation – Disaggregated Serving

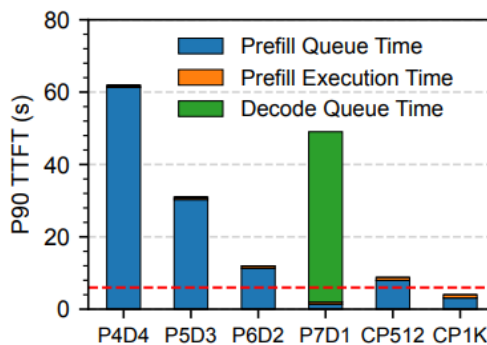
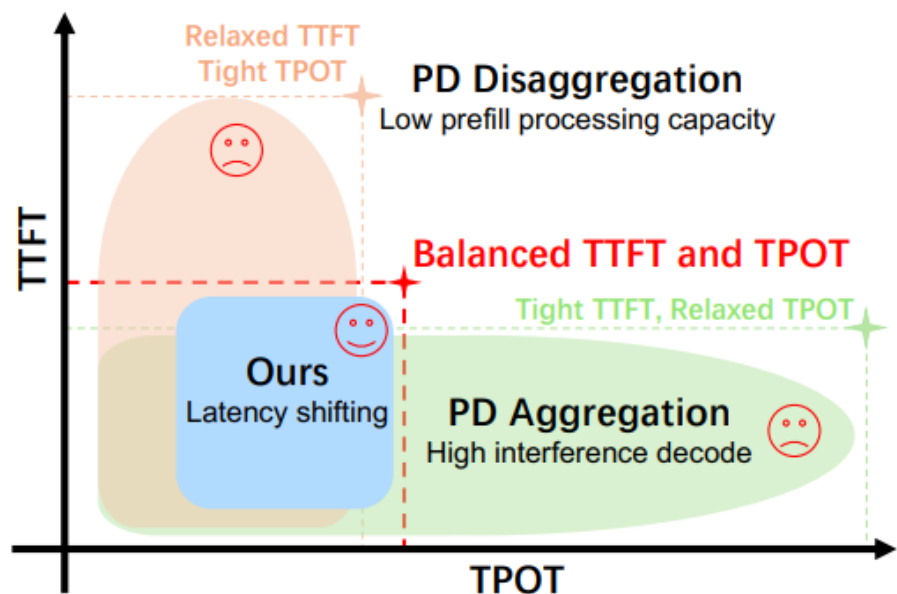


Figure 7. Breaking down the P90 tail TTFT of PD disaggregation (Px Dy) and PD aggregation (CPxxx).

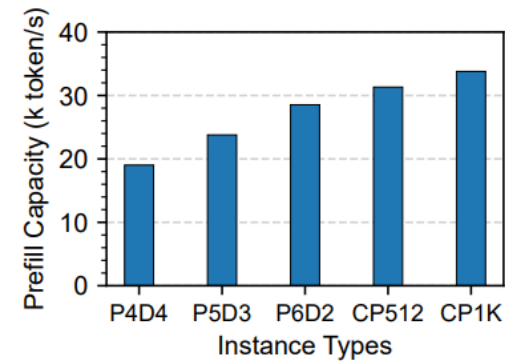


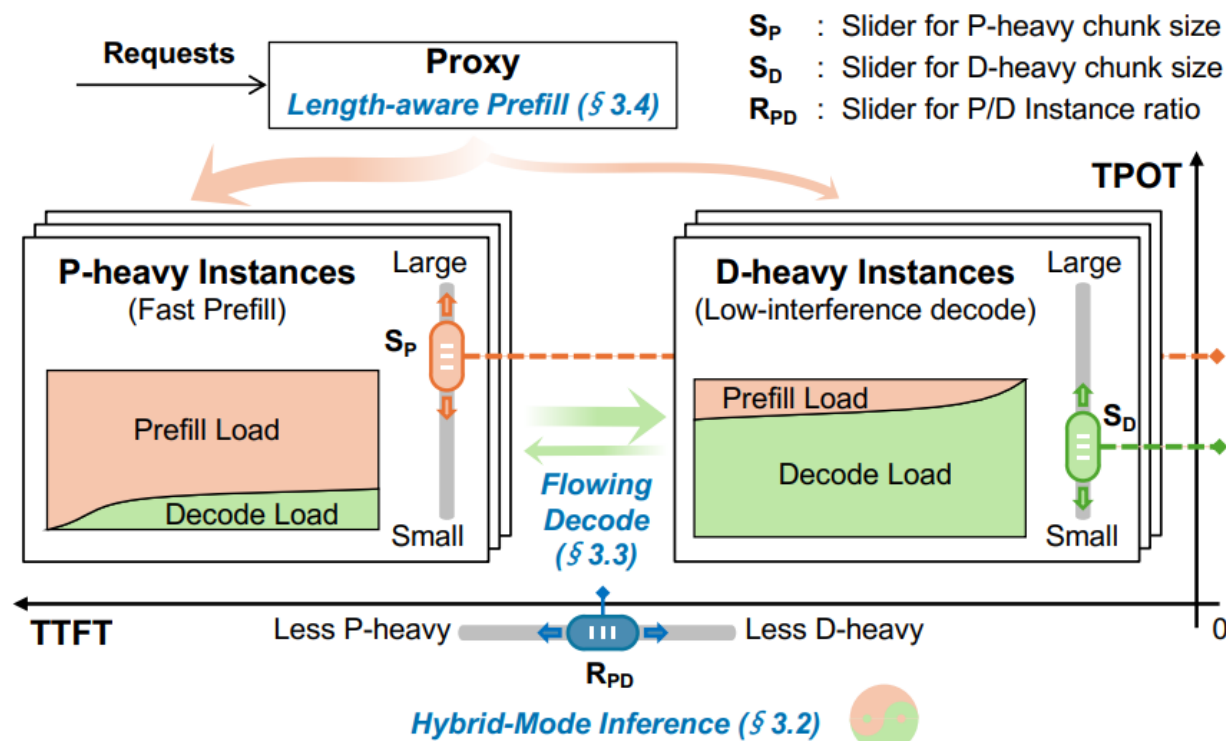
Figure 8. Prefill processing capacity of different configurations of instances.



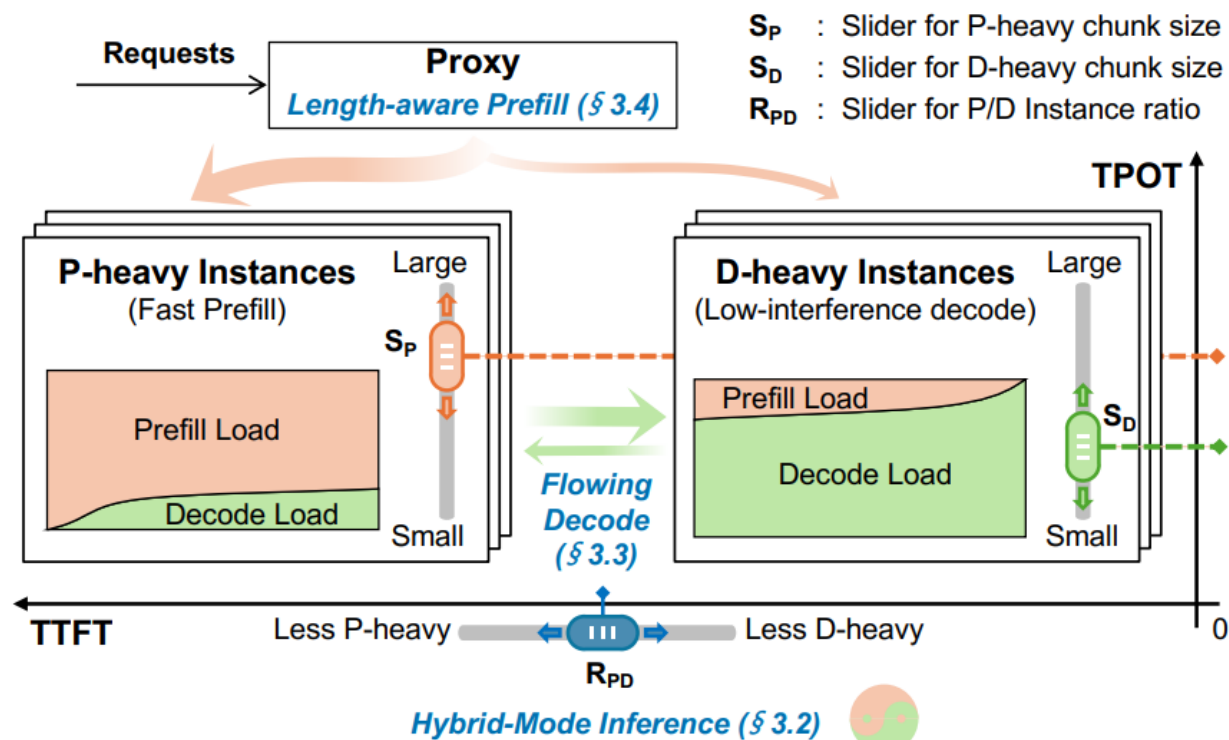
Opportunities & Challenges

- Opportunities
 - Well-satisfied latencies leave rooms for shift latencies
 - Latencies can be shifted by resource reallocations
- Challenges
 - Lack of architectural support for request-level latency shifting
 - TPOT shifting hindered by batching and unpredictability
 - TTFT shifting should consider input lengths and queue sizes

Design Overview



Design Overview



Aggregated batch handling:
Mixing P and D workloads
to reach better prefill
throughput

Disaggregated request
handling:
P and D phases can be
performed on different
instances

Decode Scheduling

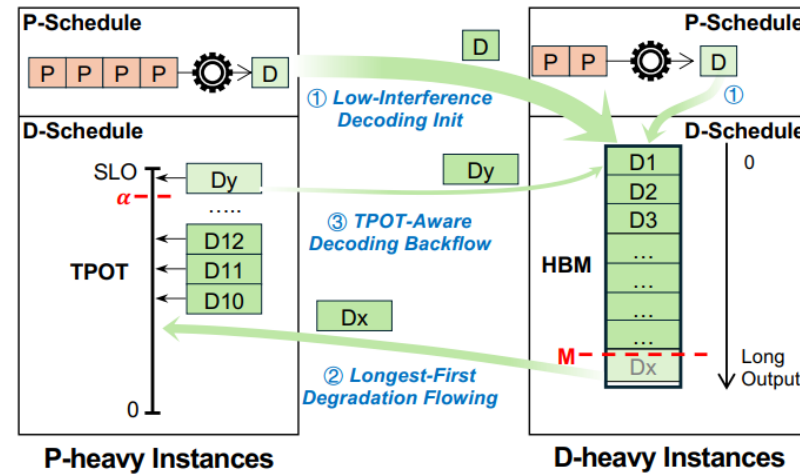


Figure 12. An illustration of flowing decode scheduling. The scheduler manages TPOT by ① starting decode requests on D-heavy instances, ② offloading the longest ones to P-heavy for TPOT degradation, and ③ returning any that approach their TPOT SLO.

Prefill Scheduling

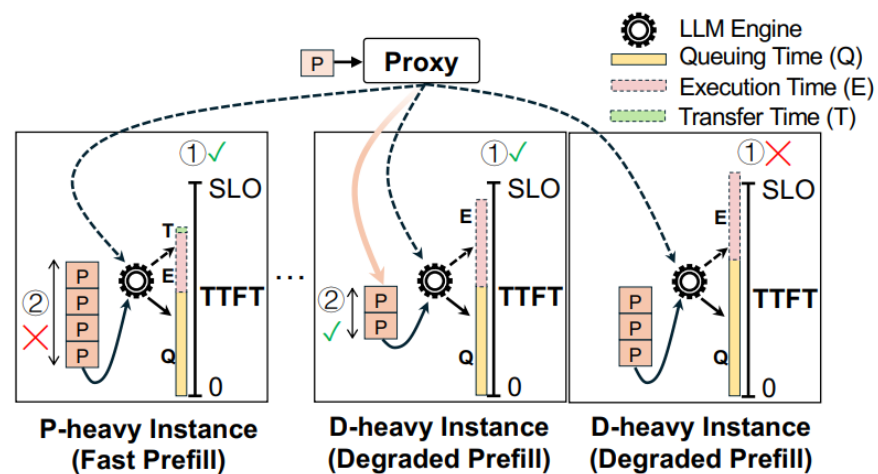


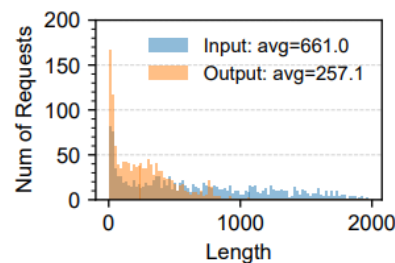
Figure 13. An illustration of length-aware prefill scheduling. The scheduler preferentially assigns short prefill requests to slower D-heavy instances when their estimated TTFT (queuing + execution + transfer time) satisfies the SLO. This reserves the faster P-heavy instances for long, more time-sensitive requests to ensure they also meet their SLOs.

Evaluation – Settings

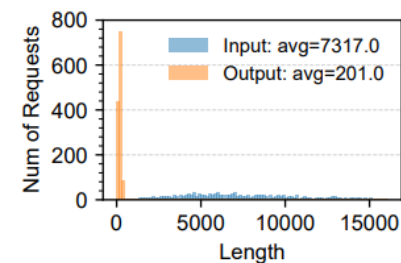
- 8 × A100 80GB
- Qwen2.5-14B and 32B (TP = 2)
- Workloads: ShareGPT, Arxiv Summarization
- Baselines: chunked prefill, disaggregation in vLLM

Table 3. Evaluated workloads and SLO constraints.

(TTFT, TPOT)	SLO1	SLO2
ShareGPT	(3s, 110ms)	(4s, 70ms)
Arxiv Summarization	(4s, 70ms)	(6s, 50ms)



(a) ShareGPT



(b) Arxiv Summarization

Evaluation – Results

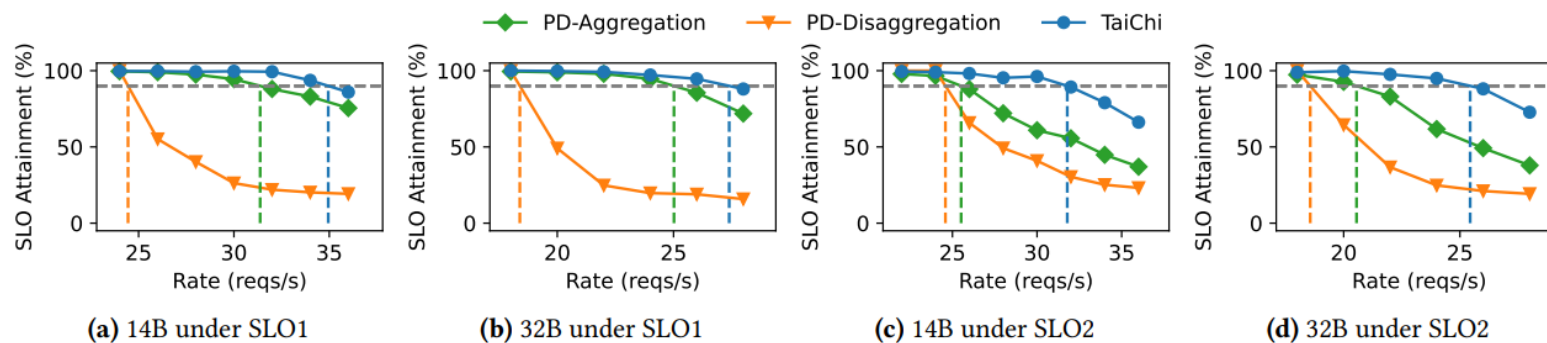


Figure 15. Goodput (vertical lines) for chatbot tasks using Qwen-2.5 models under SLO1 and SLO2.

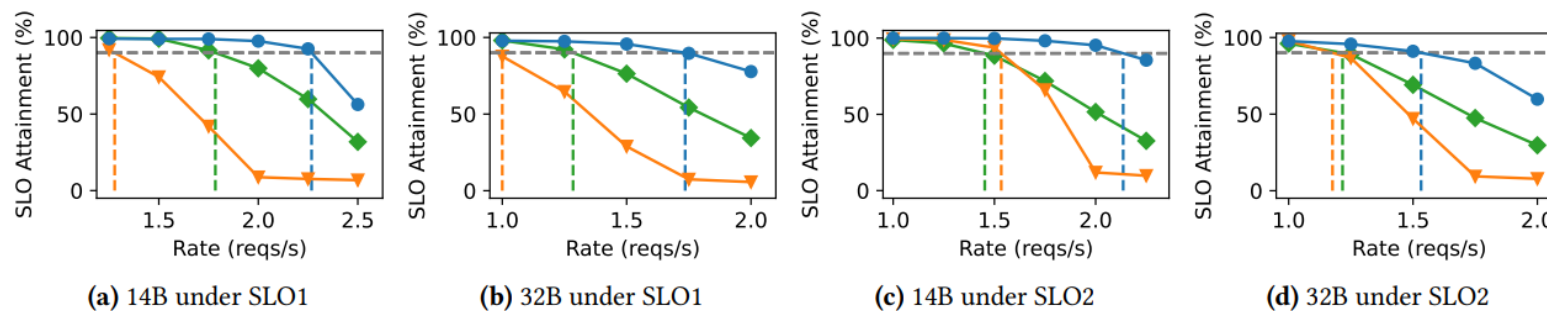


Figure 16. Goodput (vertical lines) for summarization tasks using Qwen-2.5 models under SLO1 and SLO2.



Limitations of P-D Disaggregation

- Discussion: what are your opinions?



Limitations of P-D Disaggregation

- Discussion: what are your opinions?
- Prefill
 - Less computational capacity
 - KV Cache transfer latency
- Decoding
 - Less computational and memory resources



Homework Review

- Discrepancies between DistServe and TaiChi?



Homework Review

- Discrepancies between DistServe and TaiChi?
- Whether P-D disaggregation can improve the serving quality?
- Reasons
 - Baseline
 - Model
 - Design
 - SLO



Homework Review

- Arguments about *Throughput Is Not All You Need?*
- Factors
 - Human feelings/SLOs
 - Dynamic traffic
 - Network



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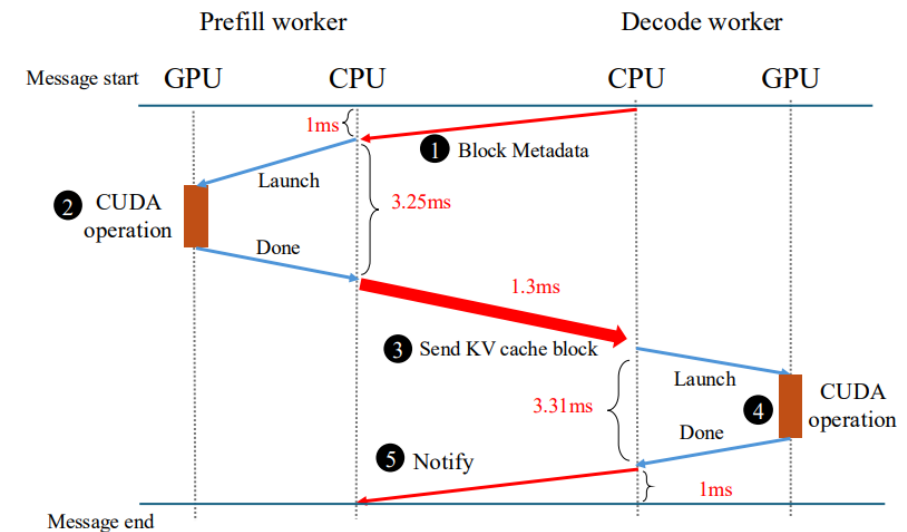


Figure 3: The message-based KV cache transfer with 4KB block size, where the blue and red arrows represent the communication over PCIe and network, respectively.

<https://arxiv.org/pdf/2501.14743>

Homework Review

- Arguments about *Throughput Is Not All You Need?*
- Factors
 - Human feelings/SLOs
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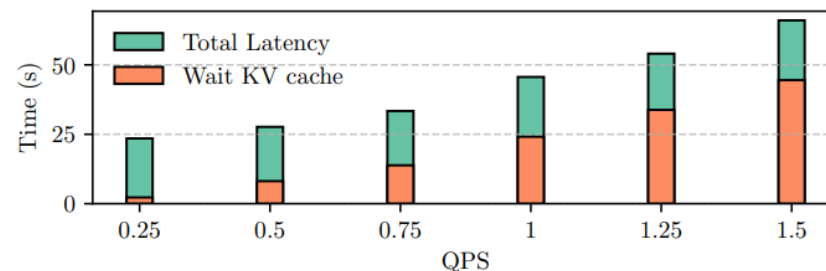


Figure 6: The latency of 16K-token requests with 70B model.

<https://arxiv.org/pdf/2501.14743>



Homework

- Read the related materials, and answer the following questions:
- 1. Why are W4A16, W4A4, and W8A8 not efficient?
- 2. How does QServe mitigate the cost of dequantization? You may answer from both QoQ quantization and QServe system.
- 3. How does COMET make one step forward in W4A4KV4 serving?
- Related materials:
- QServe paper: <https://arxiv.org/pdf/2405.04532>
- COMET paper: <https://dl.acm.org/doi/pdf/10.1145/3676641.3716252>

Q & A

